

HinglishNano-V5: A 7.7 MB Multi-Task Encoder with Deployment-Grade Retrieval Fidelity for Code-Mixed Hindi–English

Akash Yaduwanshi
aakashyaduwanshi0470@gmail.com

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Abstract

HINGLISHNANO-V5 is a 7.77M-parameter transformer (7.7 MB dynamic INT8, 9.2 ms single-thread CPU P_{50}) that produces, in a single forward pass, a 384-dimensional retrieval embedding together with memory-gating, domain, temporal-scope, and named-entity predictions for code-mixed Hindi–English (Hinglish). On a held-out full-corpus retrieval task (300 queries against 1,245 candidate memories), HINGLISHNANO INT8 retrieves the correct memory within the top-10 for 99.7% of queries—statistically indistinguishable from an **all-MiniLM-L6-v2** fine-tuned on the identical split (100.0%; $\Delta = 1/300$ queries, paired bootstrap CI $[-1.0, +0.0]$ points)—while using $2.9\times$ fewer parameters, a $2.8\times$ smaller INT8 footprint, and $1.9\times$ higher single-thread throughput (P_{50} 9.2 vs. 17.0 ms). The fine-tuned teacher retains a significant 4.0-point top-1 advantage (88.0 vs. 92.0 R@1, paired $p = 0.023$), which we report openly; a budget-matched control further shows that a shorter fine-tuning schedule (3 epochs, 195 steps) reverses the ranking entirely, a baseline pitfall we argue is common in small-model evaluations. Zero-shot general-purpose embedders collapse at corpus scale (57–67 R@1), and a 278M Hinglish-domain encoder loses 24.3 R@1 points when the candidate pool grows from 20 to 1,245—a degradation predictable from its embedding margin (+0.03). We release the held-out test split, all benchmark scripts, and per-query results.

1 Introduction

Conversational memory engines that run on personal devices must, for every utterance, answer several questions at once: should this be stored at all (gating), what is it about (domain), when does it apply (temporal scope), which entities does it mention, and—at retrieval time—which stored memory does a new query refer to. Deployed systems typically answer these with separate models, which multiplies footprint and latency. The constraint is sharpest for code-mixed Hindi–English (Hinglish), the dominant written register for hundreds of millions of users: models must handle code-mixing while remaining small enough for on-device inference, and the 384-dimensional embedding dimension aligns with widely deployed vector stores (e.g., **pgvector**).

We present HINGLISHNANO-V5, a single 5-layer, $d=384$ transformer that answers all of the above in one forward pass via five output heads, at 7.77M parameters and a 7.7 MB dynamic-INT8 ONNX artifact. The model is trained from scratch on a 24K-utterance Hinglish memory corpus with embedding-level distillation from **all-MiniLM-L6-v2** and in-batch InfoNCE.

The second contribution of this paper is evaluation rigor, which we treat as load-bearing rather than cosmetic. Small-model papers are frequently compared against zero-shot or undertrained large

baselines; both failure modes occurred in early versions of our own evaluation and were caught only by explicit controls. Our protocol therefore includes: (i) a held-out split with explicit contamination guards; (ii) canonical pooling for every baseline (BGE is CLS-pooled; mean-pooling it deflates a SOTA baseline by several points); (iii) a pinned single-thread CPU latency protocol with INT8-ONNX exports for all systems; (iv) paired bootstrap significance on identical query sets; and (v) a budget-matched fine-tuning control for the strongest baseline.

Our contributions:

- **Model.** A 7.77M-parameter, 5-head multi-task encoder for Hinglish memory retrieval; 7.7 MB INT8 at 9.2ms single-thread P_{50} , with quantization measured to be quality-free (STS-B ρ 65.33 vs. 65.30).
- **Deployment-grade fidelity.** Full-corpus R@10 of 99.7%, statistically indistinguishable from its budget-matched fine-tuned 22.7M teacher ($\Delta = 1/300$ queries), at $2.9\times$ fewer parameters and $1.9\times$ the throughput, plus four auxiliary heads at no additional forward pass.
- **Honest negative results.** The teacher retains a significant 4.0-point R@1 advantage ($p = 0.023$); our English STS-B trails dedicated embedders by 21.4 points; our tokenizer-fragmentation hypothesis for the in-domain gap was *not* supported (only $1.24\times$).
- **Robustness analysis.** Embedding margin predicts candidate-set-scaling degradation: a 278M Hinglish encoder with margin $+0.03$ loses 24.3 R@1 points from 20 to 1,245 candidates, while HINGLISHNANO (margin $+0.18$) loses 4.0.
- **A baseline pitfall, quantified.** Fine-tuning the teacher for 3 instead of 30 epochs reverses the model ranking entirely (79.0 vs. 99.0 R@1), showing how easily small-model papers can report wins against undertrained baselines.

2 Related Work

Sentence embeddings. SBERT [1] established mean-pooled transformer encoders for semantic similarity; MiniLM [2] compresses via attention distillation and is the direct teacher of `all-MiniLM-L6-v2`; BGE [3] is a strong retrieval-oriented encoder that uses CLS pooling and a query instruction for short-query-to-passage retrieval. MTEB [6] and BEIR [7] are the standard general suites; our in-domain protocol is *not* BEIR and we make no MTEB comparability claims.

Code-mixed NLP. The L3Cube line [4] provides Hinglish corpora and XLM-R-based encoders [17] (`hing-bert`, `hing-roberta`); LinCE [5] centralizes code-switching evaluation, including Hindi-English NER. We use `hing-roberta` (278M) as the domain-SOTA reference encoder and defer NER capability claims to LinCE.

Distillation and compression. TinyBERT [8] distills BERT layer-wise; we instead distill at the embedding level with cosine KD [9] plus in-batch contrastive training [10]. Architecturally we adopt GQA [11], RoPE [12], SwiGLU [13], and RMSNorm [14], and deploy via dynamic INT8 quantization [15] on ONNX Runtime.

3 HinglishNano-V5

3.1 Training

Training data is a 24,039-row Hinglish memory corpus; each row contains the raw utterance `text`, a canonical `positive_memory`, up to two curated `hard_negatives`, and labels for gating, domain,

Table 1: Architecture. The embedding table is provisioned for 20,000 tokens although the tokenizer contains 9,022; ≈ 1 MB of the INT8 artifact is unused embedding capacity, disclosed rather than trimmed in this version.

Layers / width	5 / $d=384$
Attention	GQA, 6 query : 2 KV heads
FFN	640 (SwiGLU)
Position	RoPE
Norms	RMSNorm
Embeddings	factored, rank 96, vocab 9,022 (BPE, Hinglish corpus)
<i>Heads (single forward pass)</i>	
Retrieval embedding	384 (L2-normalized)
Memory gate	2 (store / discard)
Domain	6
Temporal scope	3 (past / present / future)
NER	17 BIO tags (8 entity types)
Parameters	7.77M
ONNX size	29.83 MB FP32 / 7.70 MB INT8 (dynamic, MatMul+Gather)

temporal scope, and entities. We additionally generated 840 synthetic chatter utterances (21 strings $\times$ 40), intending to balance the gate’s discard class, and appended them after the raw corpus. Under the sequential 90/10 split this placement has a consequence we disclose rather than hide: the training split ends at raw row 22,391 of 24,039, so *all 840 chatter rows fall into the validation split*—the gate never trained on synthetic chatter (Sec. 6). The training split is therefore the first 22,391 raw rows; the final 1,648 raw rows are held out for all evaluation in this paper. Teacher targets are 384-dim normalized embeddings of `all-MiniLM-L6-v2`, cached once over all samples. The embedding loss is in-batch InfoNCE ($\tau=0.07$, batch 128) between `text` anchors and `positive_memory` positives, combined with cosine distillation ($T=4$) toward the teacher. Classification heads use cross-entropy (NER with *O*-tag down-weighted to 0.1 and entity tags up-weighted to 2.5; gate with label smoothing 0.05; domain/temporal on memory-bearing rows only). Training: 30 epochs, AdamW (weight decay 0.01), OneCycleLR (6×10^{-4}), FP16 AMP; 5,250 gradient steps.

Disclosed implementation flaws. Four details of the released checkpoint are known defects rather than hidden limitations: (i) rows lacking a resolvable `positive_memory` (26% of the corpus) fell back to using the anchor text as its own positive, contributing near-zero contrastive gradient; (ii) only the first of the two curated hard negatives was consumed; (iii) the exported checkpoint is the epoch-8 best-composite-validation model, whereas the contrastive loss kept improving through epoch 30; and (iv) the 840 synthetic chatter rows fell entirely into the validation split (above), plausibly contributing to the gate’s chatter false-positive rate (Sec. 5.6). Section 6 assesses the contribution of (i)–(iii) to the residual top-1 gap. Quantization is *dynamic* INT8 (MatMul+Gather, QUInt8); on our CPU it costs +11% latency versus FP32 and is chosen for footprint, not speed.

4 Evaluation Methodology

4.1 Protocols

Full-corpus retrieval (primary). Each of 300 seeded test queries must retrieve its gold `positive_memory` from the entire held-out memory store (1,245 candidates). This mirrors deployment (retrieve from the whole store) and de-saturates small-candidate metrics. We report R@1, R@10, MRR@10. **1-**

Table 2: Main results: full-corpus retrieval over 1,245 held-out memories (300 queries), English STS-B (zero-shot), and single-thread CPU latency in each system’s deployment configuration (INT8 ONNX unless noted). CIs in text.

Model	Par.(M)	MB	P_{50} (ms)	STS-B ρ	R@1	R@10	MRR@10	Heads
HinglishNano V5 INT8	7.77	7.7	9.2	65.3	88.0	99.7	92.5	5
HINGLISHNANO V5 FP32	7.77	29.8	8.3	65.3	88.0	99.3	92.4	5
MiniLM-L6 (zero-shot)	22.7	21.9	16.8	86.7	59.3	86.7	68.7	–
BGE-small (zero-shot)	33.4	32.3	33.5	88.9	66.7	92.3	75.8	–
Hing-RoBERTa	278	265.9	108.9	64.9	57.0	74.0	62.5	–
TinyBERT-4L	14.4	13.9	8.6	62.9	2.3	7.7	3.7	–
MiniLM-L6 (fine-tuned)	22.7	21.9	17.0	86.2	92.0	100.0	95.2	–

vs-20 (secondary). 20 candidates = gold + the dataset’s curated hard negatives (≤ 2) + seeded corpus distractors; we additionally report HardR@1 (gold ranked first among curated negatives only) and the hard-negative margin. This protocol saturates for strong models ($R@3 \geq 99$) and we treat it as diagnostic. **English STS-B.** GLUE [16] validation split ($n=1,500$), zero-shot, reporting Spearman ρ ; all models use the same setting.

4.2 Baselines and fairness

all-MiniLM-L6-v2 and bge-small-en-v1.5 are evaluated via **sentence-transformers** with their *canonical* pooling (mean and CLS respectively); BGE queries receive the retrieval instruction on the s2p task. **hing-roberta** and **TinyBERT** have no canonical embedding pooling and use masked-mean, labeled as such; TinyBERT is included as a reference encoder, not a competitor. The decisive baseline is MiniLM-FT: all-MiniLM-L6-v2 fine-tuned on *the identical training split* with the same objective family (MultipleNegativesRankingLoss, i.e., in-batch negatives, curated negatives as additional negatives). We run it at 3 epochs (195 steps) and at 30 epochs (1,950 steps) to expose the budget sensitivity of baseline conclusions; the 30-epoch model reaches 99.0 R@1 and a training loss of 0.057, i.e., saturation, so further budget cannot improve it.

4.3 Latency protocol

All latency numbers: single-thread CPU (torch and ONNX Runtime intra/inter-op threads pinned to 1), batch 1, sequence length 64, tokenization excluded, warm-up 50 followed by 3×300 timed calls, on an Intel Xeon @2.0GHz (2 physical cores). Every baseline is exported to ONNX and dynamically INT8-quantized with the same recipe as HINGLISHNANO so latency comparisons are INT8-vs-INT8; exports are verified against PyTorch outputs (max deviation 3.3×10^{-6}).

4.4 Statistics

Metric CIs are 2,000-resample bootstraps with shared resample indices across models. Head-to-head claims use paired bootstrap over identical query sets (10,000 resamples), reported as differences with 95% CIs and two-sided p -values.

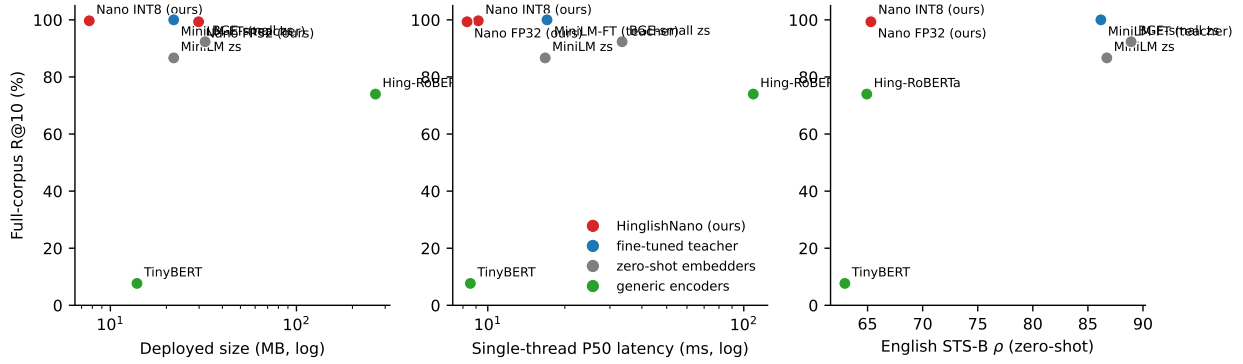


Figure 1: The quality–efficiency frontier. (a) Full-corpus R@10 vs. deployed size; (b) vs. single-thread latency; (c) the specialization trade-off: English STS-B vs. in-domain R@10. HINGLISHNANO occupies the small-fast corner without leaving the top recall tier; TinyBERT shares the corner but is not a usable embedder (R@10 7.7).

Table 3: Budget-matched control (1-vs-20 protocol). Reversal of the ranking between 3 and 30 fine-tuning epochs motivates mandatory budget disclosure for baseline fine-tuning.

System	Steps	R@1	HardR@1	hard margin	STS-B ρ
HINGLISHNANO V5 (ours, 30 ep)	5,250	92.0	92.3	+0.175	65.3
MiniLM-FT, 3 epochs	195	79.0	81.3	+0.106	86.8
MiniLM-FT, 30 epochs	1,950	99.0	99.0	+0.439	86.2

5 Results

5.1 Full-corpus retrieval

Table 2 and Figure 1 give the primary comparison. HINGLISHNANO INT8 attains $R@10 = 99.67$ (299 of 300 queries) versus the fine-tuned teacher’s 100.0: the difference is a single query (paired $\Delta = -0.33$ points, CI $[-1.00, +0.00]$, not significant; with $n=300$ we cannot resolve differences below ≈ 2 points). The teacher’s significant advantage is top-1: 92.0 vs. 88.0 (paired $\Delta = -4.0$ $[-7.3, -0.7]$, $p = 0.023$). In a memory engine whose top- k feeds downstream selection, top-10 recall is the operative metric; top-1 is the stricter proxy at which the teacher’s $2.9\times$ capacity shows.

Zero-shot embedders collapse at corpus scale (59.3–66.7 R@1) despite their STS-B strength, confirming that code-mixed in-domain retrieval requires task training. HINGLISHNANO matches the 278M Hing-RoBERTa’s 1-vs-20 HardR@1 (92.3 $[89.0, 95.3]$ vs. 93.7 $[90.7, 96.3]$, overlapping CIs) at 1/34 the INT8 size and 1/12 the latency, with $5.5\times$ its margin.

5.2 The budget-matched control

Table 3 reports the control that this paper’s claims rest on. At 3 epochs the fine-tuned teacher scores 79.0 R@1 / 81.3 HardR@1 (1-vs-20) and *loses* to HINGLISHNANO by 7 points; at 30 epochs it scores 99.0 and wins by 7. Any comparison of a task-trained small model against a lightly fine-tuned large one—in either direction—is unsound without budget disclosure. We report all HINGLISHNANO-vs-teacher statistics against the 30-epoch model.

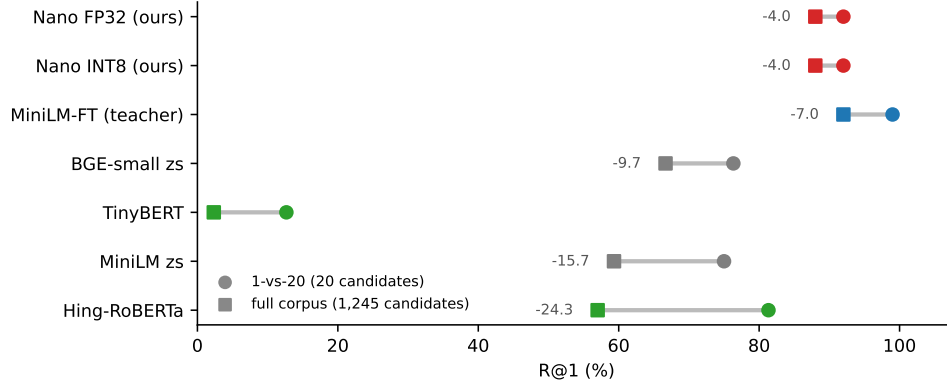


Figure 2: R@1 under candidate-set scaling (circles: 1-vs-20; squares: full corpus). HINGLISHNANO is the most scale-robust system; Hing-RoBERTa loses 24.3 points.

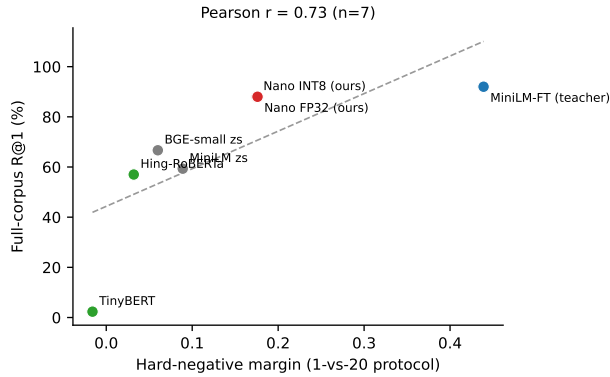


Figure 3: Hard-negative margin (small-candidate protocol) vs. full-corpus R@1. Margin predicts scale robustness; systems with near-zero or negative margins collapse when the candidate pool grows.

5.3 Robustness to candidate-set scale

Figure 2 tracks R@1 as the candidate pool grows from 20 to 1,245. HINGLISHNANO degrades least (-4.0), the saturated teacher -7.0 , BGE -9.7 , zero-shot MiniLM -15.7 , and Hing-RoBERTa -24.3 , falling *below* zero-shot MiniLM at scale. Figure 3 shows the cause: degradation is predicted by the 1-vs-20 hard-negative margin (Pearson $r = 0.73$ across the seven systems¹). Hing-RoBERTa’s compressed cosine space ($+0.03$ margin) ranks gold first among few candidates but cannot survive 1,245; margin, not R@1 on small candidate sets, is the quantity that generalizes. This is why our protocol reports both.

5.4 Latency and footprint

Figure 4. HINGLISHNANO FP32 is the fastest usable embedder measured (P_{50} 8.3 ms, 120 QPS single-thread); TinyBERT-INT8 is latency-comparable (8.6 ms) but unusable as an embedder (R@1 2.3), so we claim *fastest usable*, not fastest. HINGLISHNANO INT8 is $1.9\times$ faster than the fine-tuned teacher (9.2 vs. 17.0 ms), $3.6\times$ BGE, $11.9\times$ Hing-RoBERTa. Our own dynamic INT8 is 11% slower than our FP32 at batch 1—INT8 is the *footprint* decision (7.7 vs. 29.8 MB, $3.9\times$), FP32 the latency

¹Exact value printed by the figure script; dominated by the TinyBERT extreme.

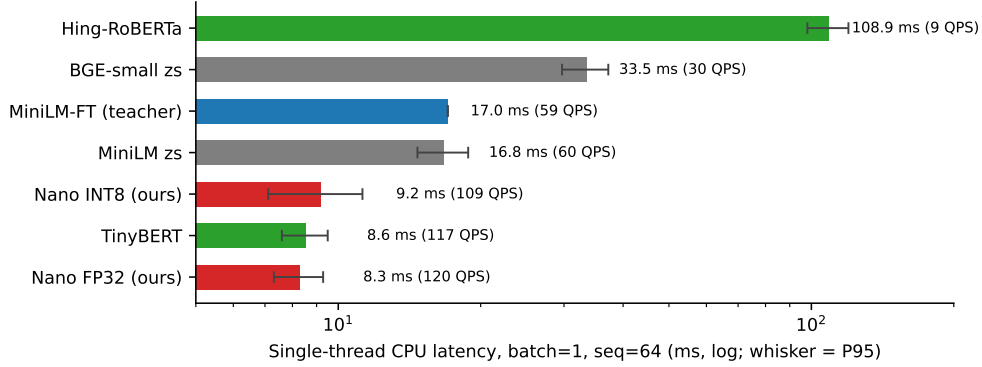


Figure 4: Single-thread CPU latency (batch 1, seq 64, tokenization excluded; whiskers = P_{95}). All systems INT8-ONNX except HINGLISHNANO FP32.

Table 4: Auxiliary heads on the 1,648-row held-out split (INT8 checkpoint). NER quality is deliberately not claimed (see text).

Gate	90.5% acc / 93.7 F1 (P 93.9, R 93.4)
Chatter false-positive probe	20.3% [95% CI 12.0, 32.3] ($n=59$)
Domain (6-way)	65.4% (weakest head)
Temporal scope (3-way)	86.7%
NER (17-tag BIO)	capability claim <i>withheld</i> : in-distribution span-F1 65.3 drops to 18.5 on out-of-template data $\Rightarrow$ template-bound deferred to LinCE <code>ner_hineng</code> evaluation

decision; both configurations are reported and quantization is quality-free (identical retrieval and STS-B to within one query and 0.03ρ).

5.5 English STS-B: the specialization trade-off

HINGLISHNANO scores $\rho = 65.3$ [62.3, 68.3], statistically indistinguishable from Hing-RoBERTa (64.9 [61.7, 67.9]) and TinyBERT (62.9), and 21.4–23.6 points below MiniLM/BGE. This is the declared cost of a 7.77M domain-specialized student; we do not claim general-purpose embedding capability. Notably, the fine-tuned teacher *retains* its English STS-B (86.2), so the in-domain gap between student and teacher is not explained by specialization damage in the teacher.

5.6 Multi-task heads (held-out)

The gate performs solidly on information-bearing utterances but suppresses only 79.7% [95% CI 67.7, 88.0] of chatter. The false positives are not random: they concentrate on *evaluative and reactive* utterances (“hilarious”, “superb”, “too funny”) and self-state reports (“thak gaya hu”), indicating the gate keys on affective rather than informational content. A structural cause is documented in Sec. 3.1: the 840 synthetic chatter rows fell entirely into the validation split, so the gate never trained on synthetic chatter of any kind. Domain is the weakest head (65.4%). NER remains unclaimed pending the LinCE evaluation.

6 Discussion

Why the student trails at top-1. Four mechanisms, in probable order of recoverability: (1) *checkpoint selection*—the deployed model is the epoch-8 composite-best, but the contrastive loss improved for 22 further epochs; (2) *signal cleanliness*—the teacher’s fine-tuning used exactly the 16,593 rows with clean (query, positive, negatives) triplets, while 26% of our contrastive batch was identity pairs; (3) *multi-task dilution*—the embedding objective shares capacity and gradient with four classification heads via uncertainty weighting; (4) *capacity* ($2.9\times$). We tested a fifth hypothesis and rejected it: the domain-native tokenizer yields only $1.24\times$ less fragmentation than MiniLM’s WordPiece (18.9 vs. 23.4 tokens/query), far too small to explain a 4-point gap. We report this negative result because tokenizer nativeness is a common but rarely quantified justification.

Deployment guidance. For a memory engine whose top- k feeds selection or a reranker, HINGLISHNANO INT8 is the Pareto choice: top-10 recall within one query of the saturated fine-tuned teacher, $2.8\times$ smaller, $1.9\times$ faster, and four decisions for free. When exact top-1 autonomous matching is required and 22 MB is affordable, the fine-tuned teacher remains better. INT8 vs. FP32 is a size-vs-latency decision, not a quality one.

7 Limitations

(1) $n=300$ test queries caps detectable differences at ≈ 2 points. (2) The store contains 1,245 memories; behavior at 10^4 – 10^5 candidates is untested, though the margin analysis suggests ordering is preserved. (3) Latency is measured on one CPU (Xeon @2.0 GHz, 2 cores); other hardware may reorder the INT8/FP32 trade-off. (4) Single training seed; inference-side CIs are reported but not training-seed variance. (5) The domain head is weak (65.4%) and the gate misclassifies evaluative chatter (Sec. 5.6). (6) NER is unevaluated on public benchmarks and prior evidence indicates template sensitivity. (7) The corpus is proprietary; the held-out test split, scripts, and per-query results are released to mitigate but not eliminate this.

8 Conclusion and Future Work

HINGLISHNANO-V5 demonstrates that a sub-8 MB multi-task encoder can hold deployment-grade retrieval fidelity—top-10 recall statistically tied with a budget-matched fine-tuned teacher $2.9\times$ its size—while paying a quantified, disclosed price in top-1 precision and general-purpose STS. Future work, in order of expected payoff: checkpoint selection by held-out retrieval metric rather than composite score; filtering identity-positive pairs from the contrastive loss; consuming both curated hard negatives; moving the chatter augmentation into the training split and extending it with evaluative and reactive utterances; trimming the embedding vocabulary; LinCE `ner_hineng` evaluation; and store-scale testing beyond 10^4 candidates.

Reproducibility

Environment: Intel Xeon @2.0 GHz (2 physical cores), 33.7 GB RAM; Python 3.12.13, PyTorch 2.10.0, ONNX Runtime 1.30.0; all inference single-thread (torch and ORT pinned). We release: the 300-query test set with golds and the 1,245-memory candidate store; the full benchmark suite (both retrieval protocols, STS-B, latency, paired statistics); per-query rankings (`benchmark_results_v3.json`); figure-generation scripts; and the model artifacts at https://github.com/unshakensoul17/HinglishNano_

Reserach_paper and Hugging Face Hub at https://huggingface.co/unsahkensoul18/Hinglishnano_V5.

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A 1-vs-20 protocol (secondary results)

B Integrity checks

The exported graph was verified to consume its attention mask (max embedding deviation 2.2×10^{-1} when the mask is forced to all-ones on padded input); ONNX exports of baselines match PyTorch outputs to 3.3×10^{-6} ; nine adversarial inputs (empty, whitespace, emoji-only, 300-character, code

Table 5: 1-vs-20 protocol: 20 candidates = gold + curated hard negatives (≤ 2) + seeded corpus distractors. Saturates for strong systems ($R@3 \geq 90.7$). 95% bootstrap CIs in brackets.

Model	R@1	R@3	MRR@10	HardR@1	margin
HINGLISHNANO V5 INT8	92.0 [88.7,95.0]	100.0	95.7	92.3 [89.0,95.3]	+0.175
HINGLISHNANO V5 FP32	92.0 [88.7,95.0]	100.0	95.7	92.3 [89.0,95.3]	+0.176
MiniLM (zs)	75.0 [69.7,80.0]	97.0	85.2	79.3 [74.7,84.0]	+0.089
BGE-small (zs)	76.3 [71.7,81.0]	98.3	86.9	78.7 [74.0,83.3]	+0.060
Hing-RoBERTa	81.3 [77.0,85.7]	90.7	86.6	93.7 [90.7,96.3]	+0.032
TinyBERT-4L	12.7 [9.0,16.7]	33.7	28.5	36.3 [31.0,42.0]	-0.016
MiniLM-FT (30ep)	99.0 [97.7,100.0]	99.0	99.5	99.0 [97.7,100.0]	+0.439

snippets, punctuation-only) produce valid output shapes with zero crashes; the dataset row count is asserted equal to the training-time count (24,039) before any split logic executes, guarding against silent split contamination.